Agentic Capabilities Orchestration

Gia, Barbara, Micheal, Gary, Lincon

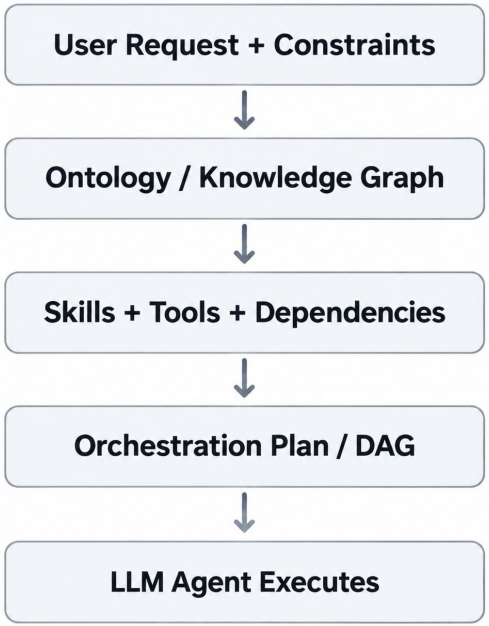
Use Case Summary

Goal:
Guide an LLM agent in choosing which skills and tools to use, in what order, and why.

Problem:
LLMs usually select tools implicitly, making orchestration hard to explain, reproduce, and validate.

Proposed solution:
Use an ontology as a semantic layer to model capabilities, inputs, outputs, dependencies, constraints, and provenance.

Use case:
Personalized supplement recommendation using PubMed, Supplement Advisor, OpenFDA, and iHerb.



Ken: affordable recommendations

USER AND CONTEXT

21-year-old college student
Trains 5–6 days per week
No known conditions or prescriptions

Goal

Lean muscle and long-term health

Two cost constraints

Supplement prices
LLM tokens and external tool calls

WHAT KEN WANTS

Affordable supplements

Strong customer reviews and good value

Lower AI execution cost

Fewer tokens and external tool calls

Scientific evidence

Accepts less analysis depth to reduce cost

No unnecessary supplements

Avoid overlapping benefits

Desired outcome: affordable, evidence-supported choices with fewer unnecessary calls

Jessica: safety and evidence

USER AND CONTEXT

Late 60s, with osteoarthritis
Declining bone density
Takes celecoxib 200 mg daily

Goal

Bone health and healthy aging

Using the recommender

Access through a paid LLM
Willing to spend on quality

WHAT JESSICA WANTS

Thorough safety evaluation

Check medication interactions before recommending

High-quality scientific evidence

Support recommendations with research

High-quality supplements

Minimize the risk of adverse effects

Quality takes priority over cost

LLM usage and product prices matter less

Desired outcome: quality supplements backed by evidence and safety checks

Explanations users want

Both users inspect plans through the agent harness.

KEN

Why did the system select this plan?

He compares candidate plans and wants to understand the tradeoffs.

Selection criteria and their priority
How the system handled tradeoffs
Reasons for eliminating other plans

JESSICA

Why these capabilities, in this order?

She reviews the executed plan and wants to understand each step.

Purpose of each capability
Tools, servers and intermediate outputs
Reasons for the execution order

Both users want clear reasons for the system's decisions.

Competency Question

Q: What skills are required to satisfy the user's request?

Jessica: Provide me with supplement recommendations that suit my health profile.

A: Personalized Supplement Recommendation Skill, Supplement Safety Screening Skill, Evidence Verification Skill, Product Selection Skill

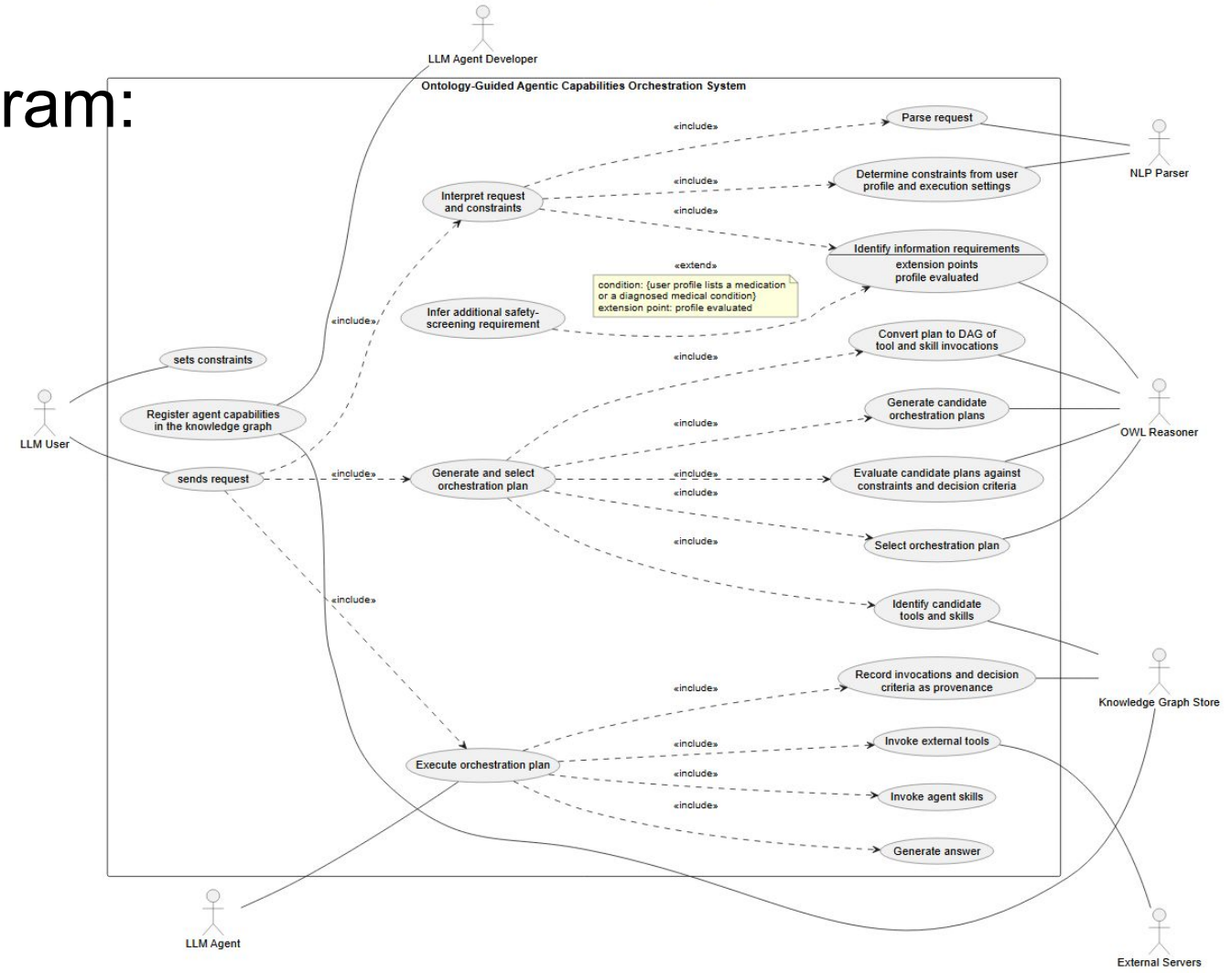
Jessica: Are there any safety concerns with taking magnesium while I'm taking my current medications?

A: Supplement Safety Screening Skill, Evidence Verification Skill

How to answer with the ontology

1. Process the user query with an NLP tool to extract the **InformationRequirements** to satisfy the user's request
2. Determine the **Constraints** from the user's health profile and execution settings and additional **InformationRequirements** the user's **OrchestrationPlan** must follow (ex. Jessica's current medication and diagnosed health condition means the OrchestrationPlan must include safety screening for contraindications)
3. Use the KG to map what **Capabilities** satisfy the **InformationRequirements** and the **Skills** that perform those **Capabilities**.
4. Return the **Skills** needed to execute in the **OrchestrationPlan**

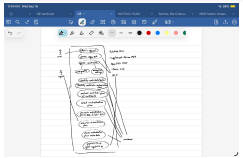
Use Case Diagram: Overall System Architecture



AI Use 1/4 — ChatGPT: the use case diagram

September 16 · ChatGPT Free model

1 WE GAVE IT

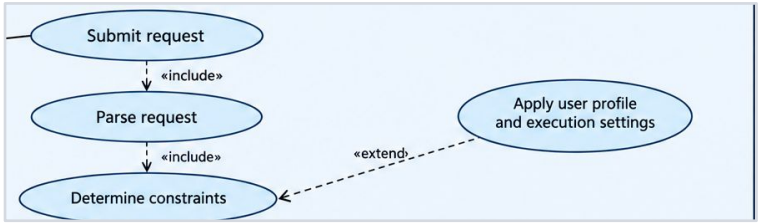


"this is a sketch of my UML use case diagram of the basic flow of events. i need this to be UML 2.5.1 compliant" + the photo

"… read page 683. it shows you what a use case diagram that complies with UML 2.5.1 format looks like … revise the use case diagram once you read it"

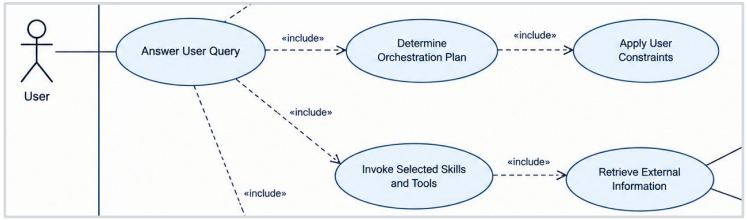
2 IT RETURNED

DETAIL — FIRST DRAFT, FROM THE SKETCH



«include» chained as a process flow; «extend» with no condition

DETAIL — AFTER WE CITED CLAUSE 18



One base use case, «include» to each; the agent and the ontology use cases are gone

"Most importantly, I removed the process-flow arrows. UML 2.5.1 explicitly treats use cases as specifications of behavior offered by a subject."

3 WE CHANGED

Carried into Figure 2: the supporting systems drawn as actors outside the subject, and the use case names.

Left out: the internal component boxes, and the revision's smaller set of use cases. We redrew the figure in PlantUML, because an image has no source to edit.

AI Use 2/4 — Claude: diagram source we could edit

September 17–18, Claude Fable 5.1 · output was PlantUML text, rendered with PlantUML and Graphviz

1

WE GAVE IT

“generate the diagrams. You should read all the referenced materials the ontology course specifies, and do the best creating the diagram.”

It read: our use case document · UML 2.5.1 clauses 11, 15, 16 and 18 · the instructors’ example use cases

2

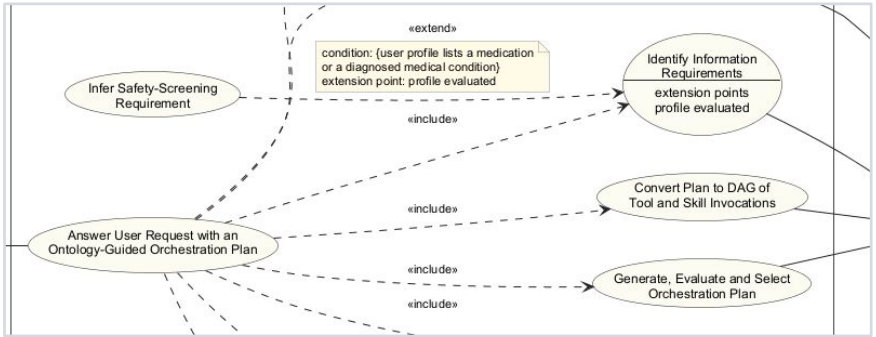
IT RETURNED

PLANTUML SOURCE, FIG2_USECASE.PUML

```
usecase "Identify Information
  Requirements
  --
  extension points
  profile evaluated" as UC2

UC0 ..> UC2 : <<include>>
UCX ..> UC2 : <<extend>>
note on link
  condition: {user profile lists
  a medication or a diagnosed
  medical condition}
  extension point: profile evaluated
end note
```

DETAIL — THE SAME LINES, RENDERED



PlantUML parses the file, reports syntax errors, and renders it. Seven files came back; each one opens in the online editor.

3

WE CHANGED

Figure 2. Added the LLM Agent Developer actor and the user’s own use cases (sends request, sets constraints); collapsed the four MCP servers and the CLI into one External Servers actor; split plan generation into generate → evaluate → select.

Figures 3 and 4. Renamed the swimlanes to NLP Tool, harness, reasoner, knowledge graph and external servers, and matched them to the eleven-step basic flow and the seven-step alternate flow.

Set aside. The architecture component diagram and the three expansion diagrams.

AI Use 3/4 — ChatGPT: alternate flow and scenarios

September 18 · ChatGPT Free model

	Alternate flow	Usage scenarios
1 WE GAVE IT	"this is the basic flow of events of my use case. help me come up with ideas for alternate flow of events, particularly error states" + our basic flow of events table, pasted in	"this is a draft of a usage scenario i was thinking adding that emphasized the orchestration use case. please revise … for clarity" + a scenario we had written ourselves
2 IT RETURNED	Fourteen flows, A1–A14, grouped as request, planning, execution and answer errors. A4: **A4 — No candidate capabilities** 5b Determines that no available tools or skills satisfy one or more information requirements 5c Identifies the unsatisfied information requirements 5f Informs the user that the available capabilities cannot satisfy the request	Ten scenario ideas, then this rewrite of our draft: **Orchestration Plan Selection and Explanation** *"Ken is able to view the candidate orchestration plans considered by the agent through the system's harness. He knows that he constrained the execution settings to minimize token usage and external tool calls … and wants to understand why a particular orchestration plan was selected."*
3 WE CHANGED	Took A4 and rewrote it as a seven-step table: our own actors (NLP Tool, harness, reasoner, agent), steps 1–3 of the basic flow in front, and no partial-fulfilment branch. **→ Section V, and Figure 4**	Kept its wording for both drafts, dropped its closing sentence, and recast the ending as what the user wants to know. **→ Section III, scenarios 3 and 4**

AI Use 4/4 — Compliance, tooling, experience

Were the generated diagrams UML 2.5.1 compliant?	ChatGPT's first draft chained «include» as a process flow and used «extend» with no condition. After we cited clause 18 it returned compliant notation and removed the flow arrows. Claude's PlantUML drafts used «include» for containment, one «extend» with its condition and extension point, swimlanes and object nodes.
Could they be generated inside a UML tool?	ChatGPT returned PNG images, so there was nothing to load. Claude returned PlantUML source: PlantUML parses it, reports syntax errors and renders it, and the three submitted figures were edited and re-rendered there. Each figure links to its source in the PlantUML online editor.
What was the experience?	Image output made every correction a full regeneration, and small errors rode along: the sequence diagram carries a step named getUserProfileAndAndSettings(). Diagram source let us rename one actor and re-render in seconds, which is why the submitted figures are PlantUML.

Lesson. Ask for diagram source rather than a picture, and cite the clause you want it to follow.

Models. ChatGPT desktop (OpenAI; the app shows no version) and Claude Fable 5.1 (Anthropic) in Claude Code. The three ChatGPT chats are linked in OE_3_TeamAgenticCapabilities_AIUse.docx.